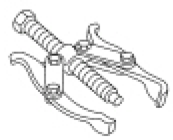
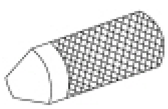
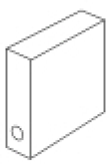
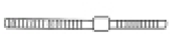
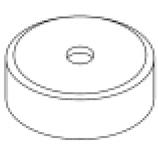
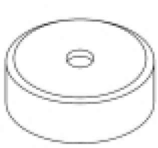
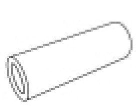
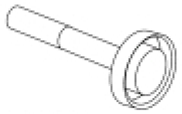
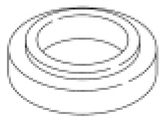
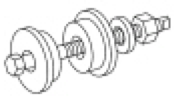
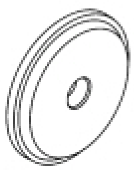
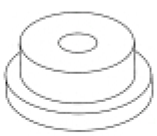
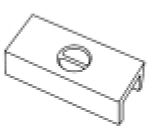
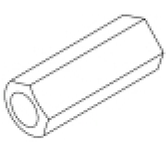
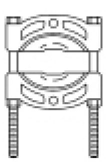
Differential Ring And Pinion

Ford Ecat Electronic Parts Catalog

Special Tool(s)

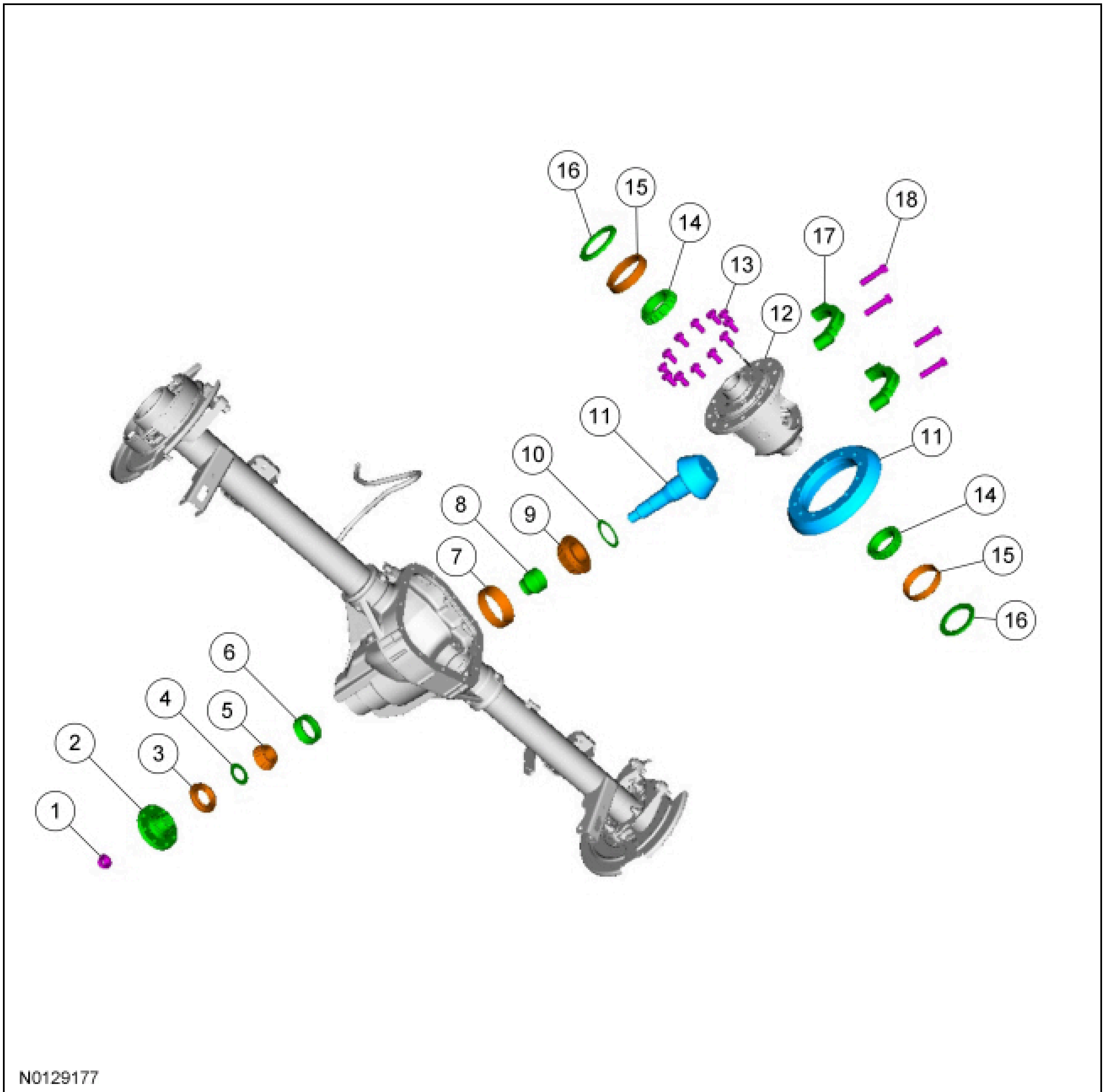
 ST2026-A	2 Jaw Puller 205-D072 (D97L-4221-A) or equivalent
 ST1432-A	Adapter for 205-S127 205-111 (T76P-4020-A11)
 ST2468-A	Adapter for 205-S156 205-159 (T80T-4020-F42)
 ST1429-A	Adapter for 205-S156 205-160 (T80T-4020-F43)
 ST1743-A	Depth Gauge/Aligner, Drive Pinion 205-937 (TKIT-2009C-F)
 ST1743-A	Depth Gauge/Aligner, Drive Pinion 205-383 (T97T-4020-B)
 ST1434-A	Gauge Tube, Drive Pinion 205-377 (T97T-4020-A)

 ST2473-A	Installer, Differential Carrier Bearing 205-D044 (D81T-4221-A) or equivalent
 ST1367-A	Installer, Drive Pinion Bearing Cone 205-488
 ST1676-A	Installer, Drive Pinion Bearing Cup 205-024 (T67P-4616-A)
 ST2368-A	Installer, Input Shaft Bearing Cup 308-391
 ST1741-A	Installer, Rear Axle Pinion Bearing Cup 205-852
 ST1254-A	Plate, Bearing/Oil Seal 205-090 (T75L-1165-B)
 ST1744-A	Protector, Drive Pinion Thread 205-229 (T85T-4209-AH)
 ST1368-A	Puller, Bearing 205-D064 (D84L-1123-A)
 ST1725-A	Step Plate 205-D061 (D83T-4205-C2) or equivalent

Material

Item	**Specification**

Maximum Strength Retaining Compound Loctite® 638™	—
Motorcraft® Premium Long-Life Grease XG-1-E1	ESA-M1C75-B



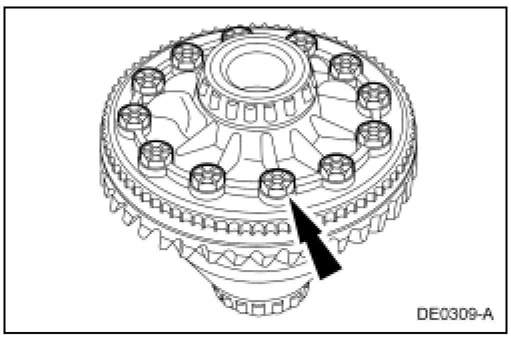
Item	Part Number	Description
1	4C1214C121	Drive pinion nut (part of 4320)
2	48514851	Drive pinion flange
3	46764676	Drive pinion oil seal
4	46704670	Pinion oil slinger
5	46214621	Outer pinion bearing
6	—	Outer pinion bearing cup (part of 4621)
7	—	Inner pinion bearing cup (part of 4630)
8	46624662	Collapsible spacer (part of 4320)
9	46304630	Inner pinion bearing
10	46634663	Pinion bearing adjustment shim

11	42094209	Drive pinion and ring gear
12	42044204	Differential assembly
13	42164216	Differential ring gear bolt (12 required)
14	42204220	Differential carrier bearings (2 required)
15	—	Differential carrier bearing cups (part of 4220)
16	40674067	Differential carrier bearing shims (2 required)
17	—	Differential bearing cap (part of 4010) (2 required)
18	4610846108	Differential bearing cap bolts (part of 4010) (4 required)

Removal

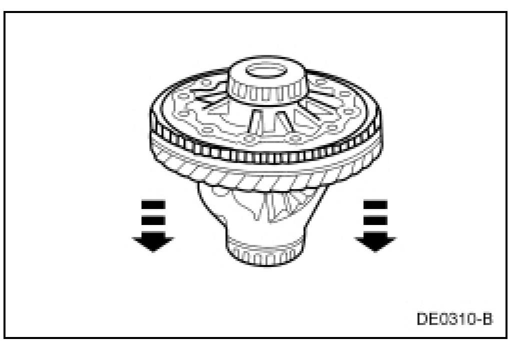
All vehicles

1. Remove the drive pinion flange seal. Refer to Drive Pinion Flange and Drive Pinion Seal.
2. Remove the differential carrier assembly. Refer to Differential Carrier.
3. Remove and discard the 12 ring gear bolts.



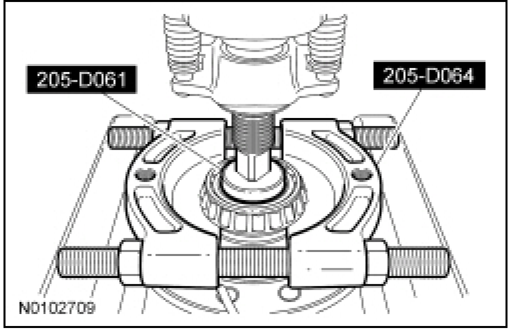
4. **NOTE:** *Do not damage the ring gear bolt hole threads.*

   Insert a punch in the ring gear bolt holes and drive the differential ring gear off.

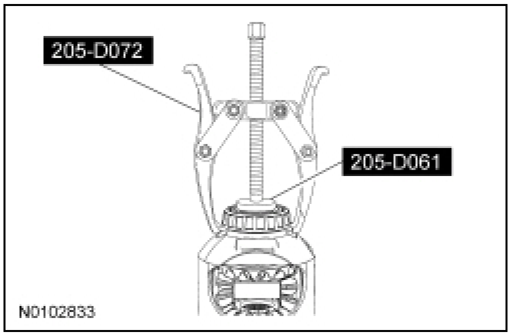


Electronic Locking Differential (ELD) equipped vehicles

5. Using the Bearing Puller plate, Step Plate and a suitable press, remove the LH differential bearing.

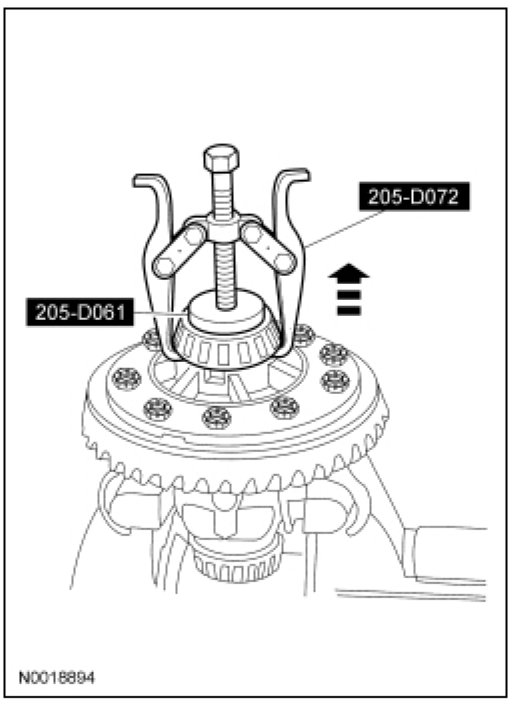


6. Using the 2 Jaw Puller and Step Plate, remove the RH differential bearing.



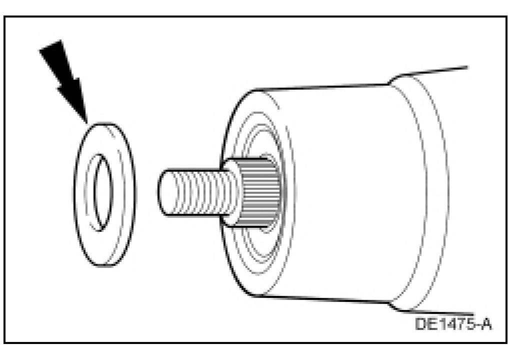
Non-ELD equipped vehicles

7. Remove the 2 differential bearings with the 2 Jaw Puller and Step Plate.

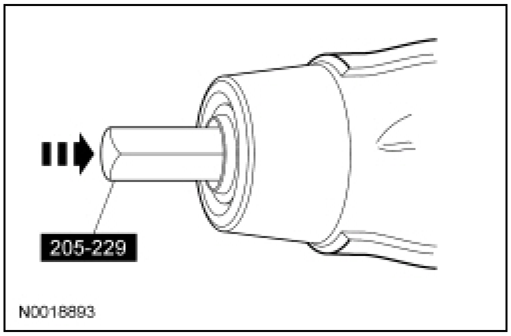


All Vehicles

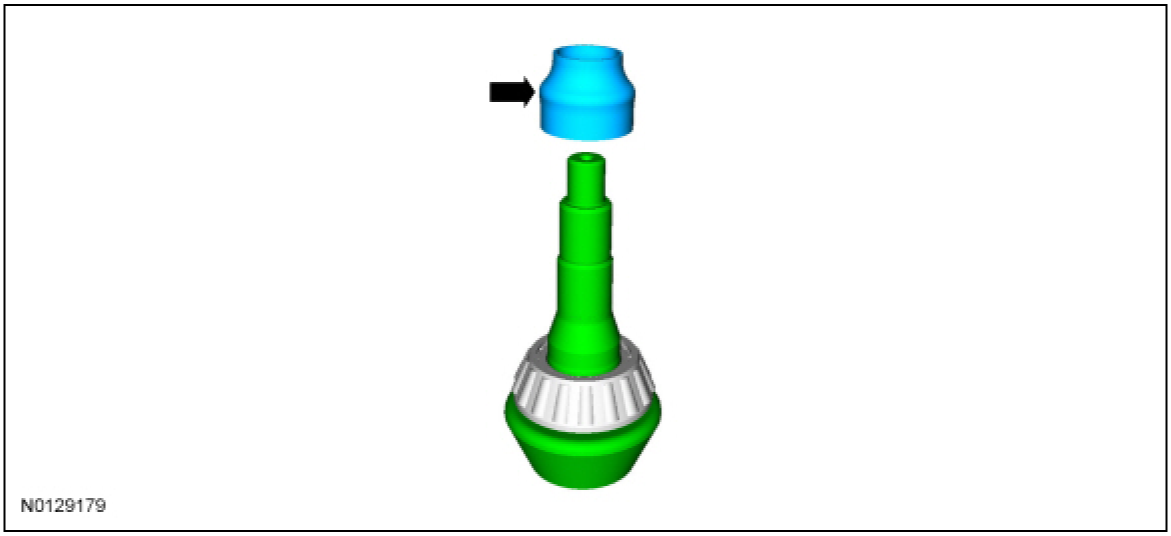
8. Remove the drive pinion shaft oil slinger.



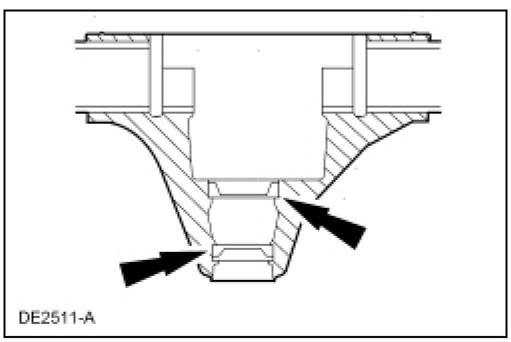
9. Install the Drive Pinion Thread Protector, then use a soft-faced hammer to drive the pinion gear out of the outer drive pinion bearing and remove the pinion gear.



10. Remove and discard the drive pinion collapsible spacer.



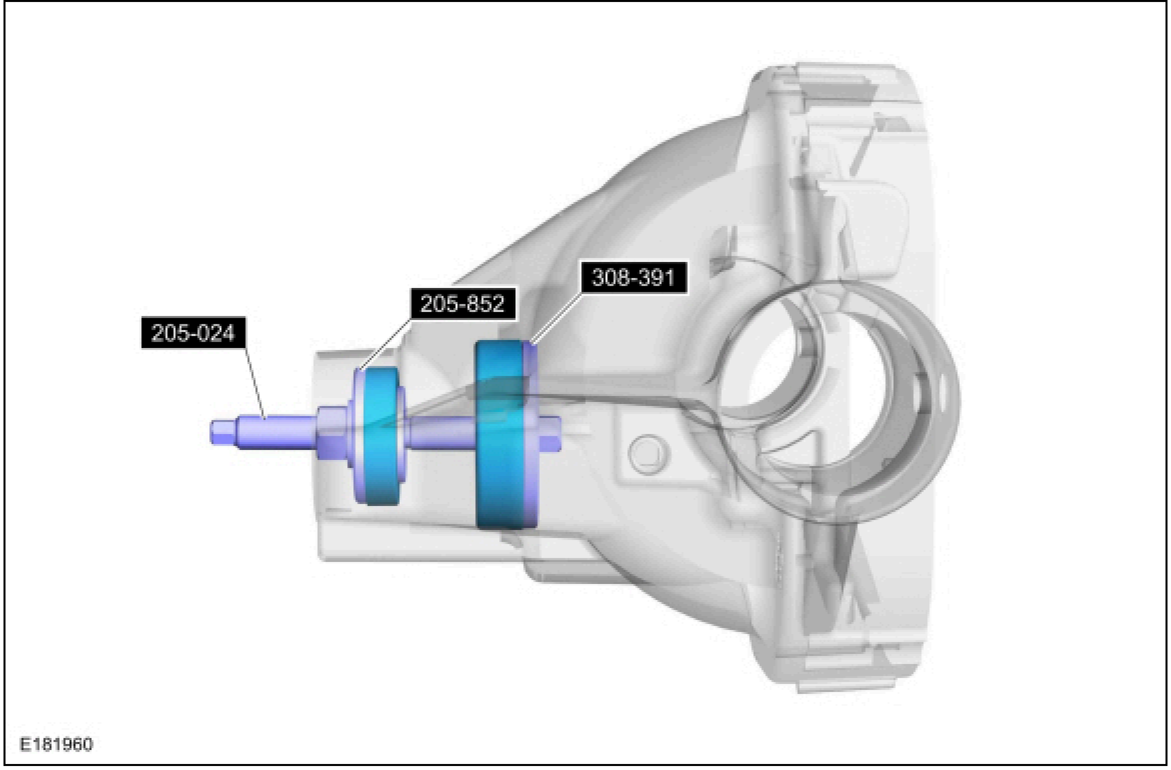
11. Using a brass drift, remove the 2 drive pinion bearing cups by tapping alternately on opposite sides of the 2 drive pinion bearing cups.



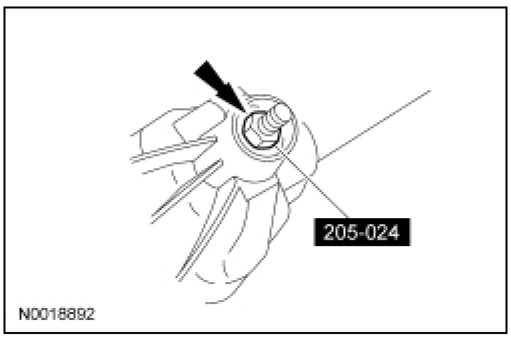
Installation

All vehicles

1. Position the Drive Pinion Bearing Cup Installer and the inner and outer drive pinion bearing cups in their respective bores.
    1. Place the Rear Axle Pinion Bearing Cup Installer on the outer drive pinion bearing cup.
    2. Place the Rear Axle Pinion Bearing Cup Installer on the inner drive pinion bearing cup.
    3. Install the Drive Pinion Bearing Cup Installer.

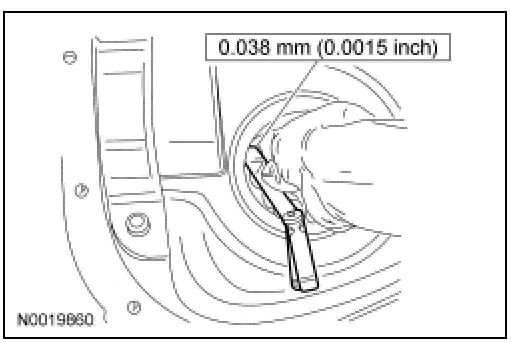


2. Tighten the Drive Pinion Bearing Cup Installer to fully seat the 2 differential drive pinion bearing cups.



3. **NOTE:** *If a feeler gauge of the specification shown can be inserted between a cup and the bottom of its bore at any point around the cup, the cup is not correctly seated.*

   Make sure the differential pinion bearing cups are correctly seated.



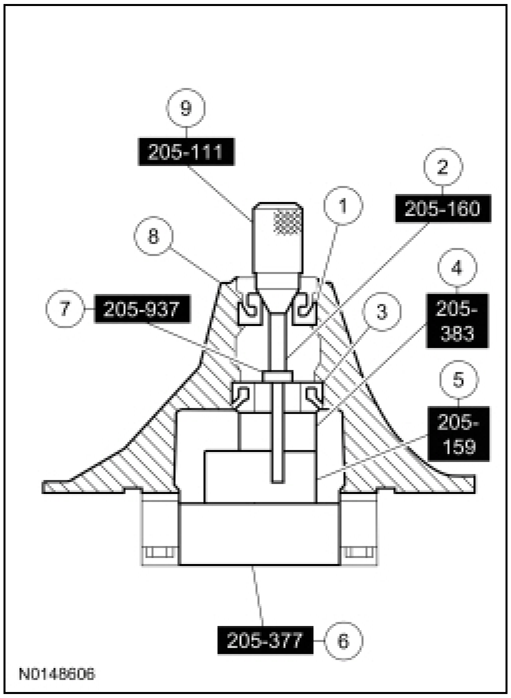
4. **NOTICE: Use the same drive pinion bearings and the drive pinion bearing adjustment shim from the drive pinion bearing adjustment shim selection procedure for final assembly or damage to the component may occur.**

   NOTE: *Install new drive pinion bearings without any additional lubricant since the anti-rust oil provides adequate lubricant without upsetting the drive pinion bearing preload settings.*

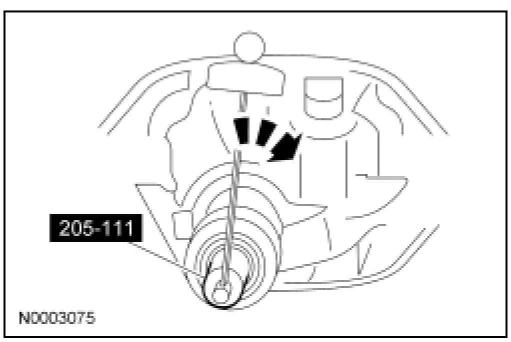
   Install the new drive pinion bearings and Adapters as shown.

Item	Part Number	Description
1	46214621	Outer pinion bearing

2	205205-160-160	Adapter for 205-S127
3	46304630	Inner pinion bearing
4	205205-383-383	Depth Gauge/Aligner, Drive Pinion
5	205205-159-159	Adapter for 205-S127
6	205205-377-377	Drive Pinion Gauge Tube
7	205205-937-937	Depth Gauge/Aligner, Drive Pinion
8	40104010	Rear axle housing outer (front) pinion bearing
9	205205-111-111	Adapter for 205-S127

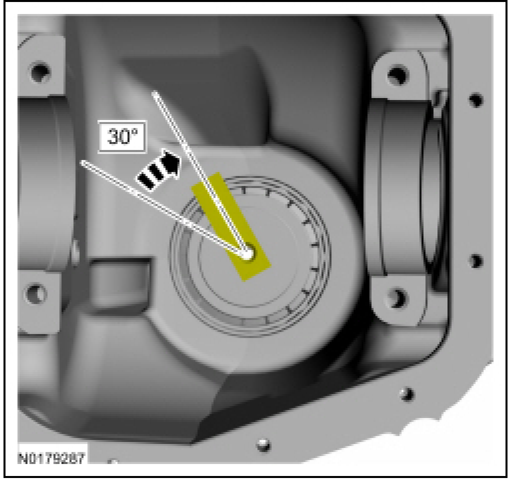


5. Tighten the Adapter to the specified rotational torque.
    - Tighten to 2.2 Nm (20 lb-in).

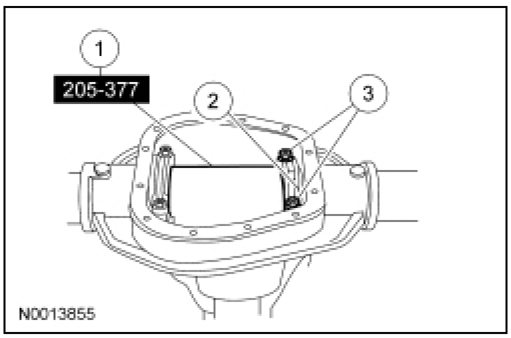


6. **NOTE:** *The Adapter must be offset to obtain an accurate reading*.

    Rotate the Adapter several half-turns to make sure of correct seating of the drive pinion bearings.



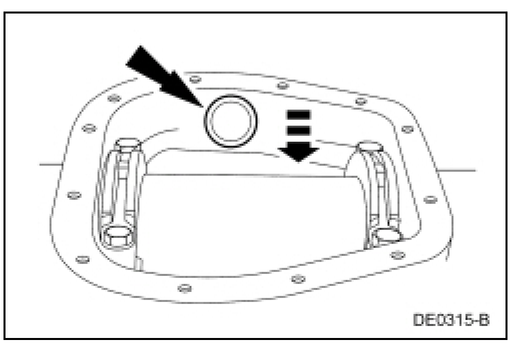
7. Install the Drive Pinion Gauge Tube.
    1. Position the Drive Pinion Gauge Tube.
    2. Install the 2 differential bearing caps.
    3. Install the 4 differential bearing cap bolts.
        - Tighten to 105 Nm (77 lb-ft).



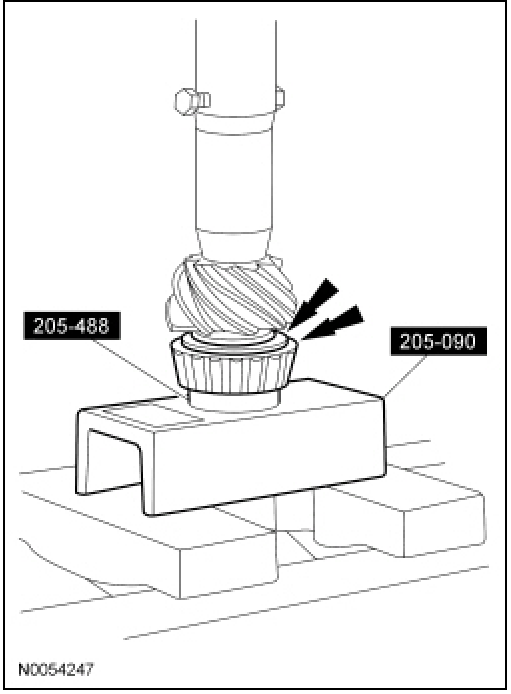
8. **NOTE:** *Drive pinion bearing adjustment shims must be flat and clean.*

    NOTE: *A slight drag should be felt for correct drive pinion bearing adjustment shim selection. Do not attempt to force the drive pinion bearing adjustment shim between the gauge block and the gauge tube. This will minimize selection of a drive pinion bearing adjustment shim thicker than required, which results in a deep tooth contact in the final assembly of integral axle assemblies.*

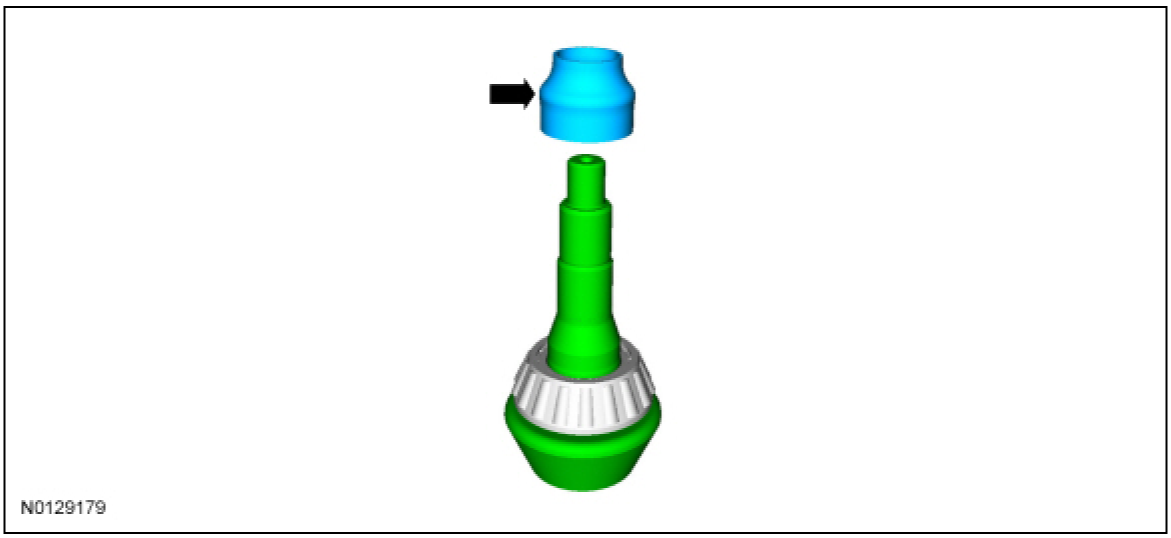
    Use a drive pinion bearing adjustment shim as a gauge for drive pinion bearing adjustment shim selection.
    - After the correct drive pinion bearing adjustment shim thickness has been determined, remove the Adapters.



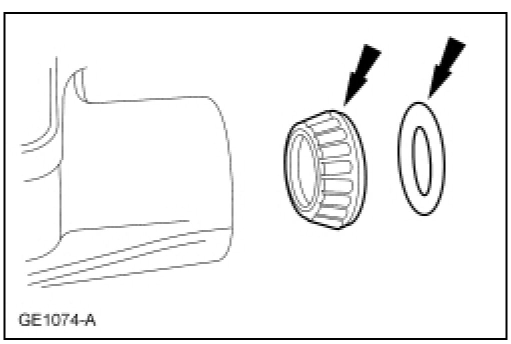
9. Using the Bearing/Oil Seal Plate and Drive Pinion Bearing Cone Installer, press the drive pinion bearing and selected drive pinion bearing adjustment shim until they are firmly seated on the pinion gear.



10. Place a new drive pinion collapsible spacer on the pinion gear against the pinion gear shoulder.

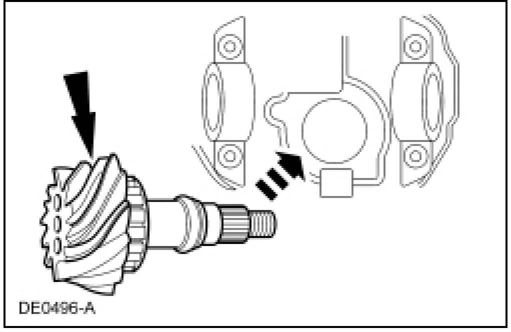


11. Install the outer drive pinion bearing and drive pinion shaft oil slinger.



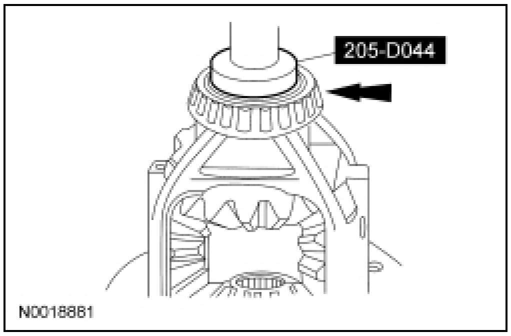
12. Lubricate the pinion flange splines with grease.

13. Install the pinion gear in the pinion gear bore from inside the axle housing.



14. Install the drive pinion seal and flange. Refer to Drive Pinion Flange and Drive Pinion Seal.

15. Using the Differential Carrier Bearing Installer, press the LH and RH differential bearing on the differential assembly.

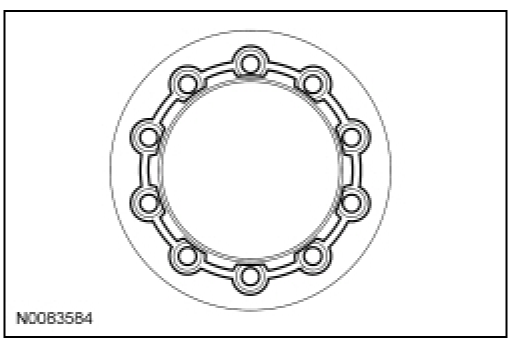


Electronic Locking Differential (ELD) equipped vehicles and all vehicles equipped with 3.55, 3.73, and 4.10 ratio gear sets

16. **NOTE:** *The differential flange and ring gear flange must be free of any old retaining compound material. Make sure both surfaces are clean and free of oil, dust and debris. Failure to clean the surfaces can result in ring gear runout concerns.*

    Clean all traces of the old retaining compound material from the differential flange.
    - Use solvent and Scotch-Brite® pads to remove.

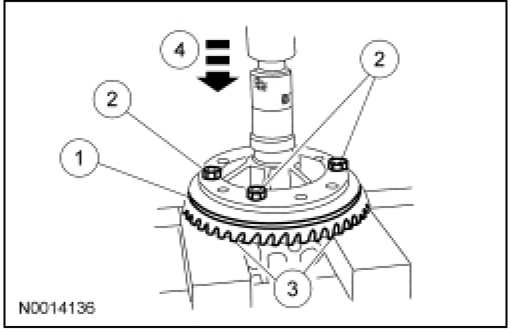
17. Apply a one-eighth inch bead of maximum strength retaining compound on the rear face of the ring gear in the pattern shown.



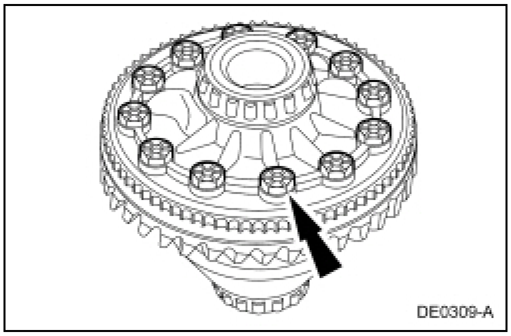
All vehicles

18. **NOTE:** *If removed, press the speed sensor ring on with the differential ring gear.*

    Install the differential ring gear.
    1. Place the differential ring gear on the differential assembly.
    2. Hand-start 3 new ring gear bolts to align the holes in the differential ring gear and the differential assembly.
    3. Place the differential assembly and differential ring gear onto the press bed blocks with the differential ring gear teeth facing downward.
    4. Press the differential ring gear into place.



19. Install the remaining 9 new differential ring gear bolts.
    - Stage 1: Tighten to 60 Nm (44 lb-ft).
    - Stage 2: Tighten an additional 90 degrees.



20. Install the differential carrier. Refer to Differential Carrier.